# CHAPTER 4

UniConnect, for Student Software Engineering Projects

**By**

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## Introduction

The UniConnect mobile application is an application designed to help students manage projects, tasks, meet deadlines and communicate with their team members all in one place.

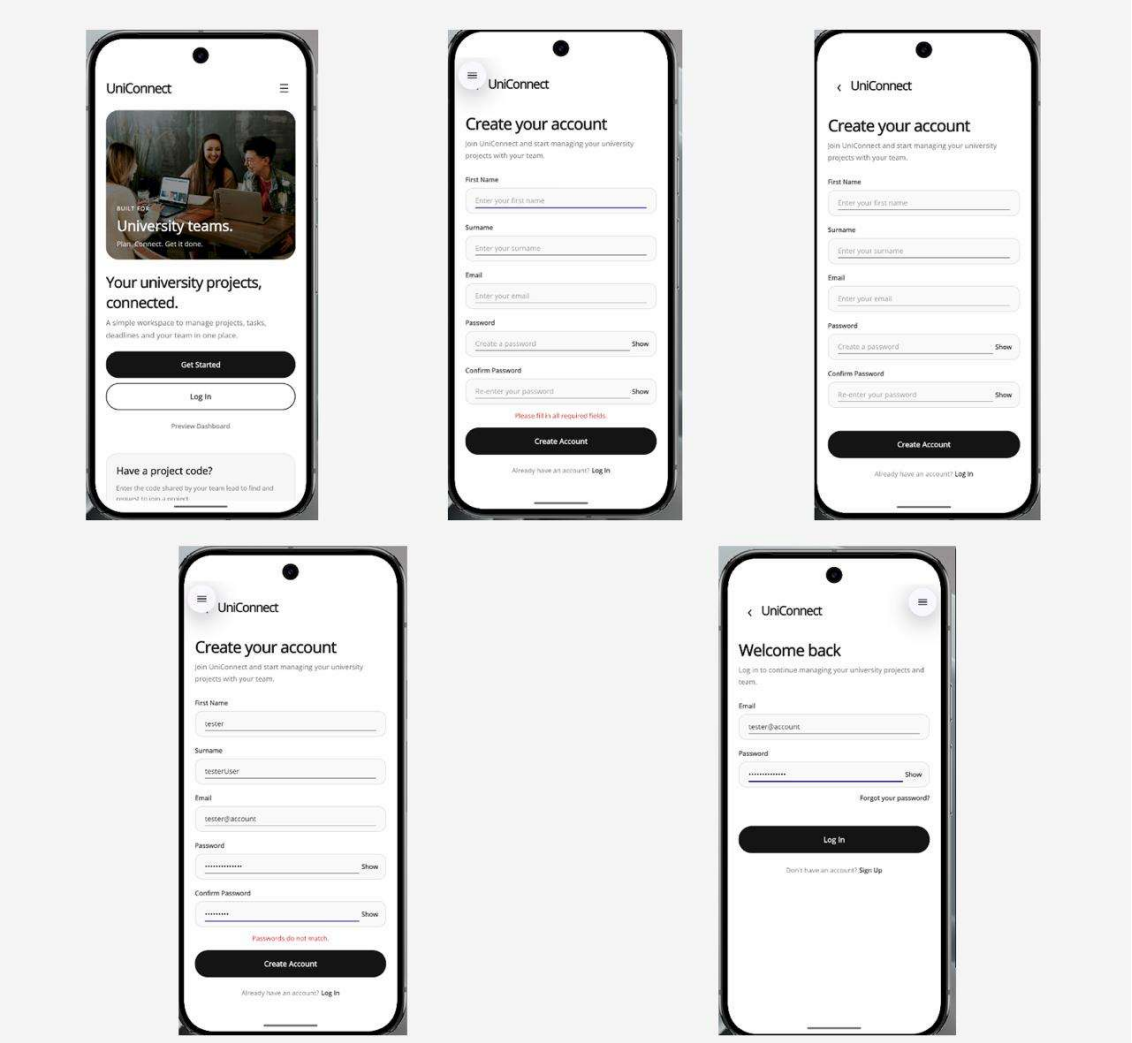
In this documentation this deliverable will focus on illustrating how using .NET MAUI, C#, Java Spring Boot and MySQL came together and were able to make up this application. The document will have screenshots of the developed screens and evidence of the application's functionality, navigation, backened integration and frontend.

## The System Design:

UniConnect is essentially built on .NET MAUI for the frontend and Spring Boot and Java for the backend. The system's design will be shown in the screenshots below where we disect the system from the top down.

## The Authentication Layer

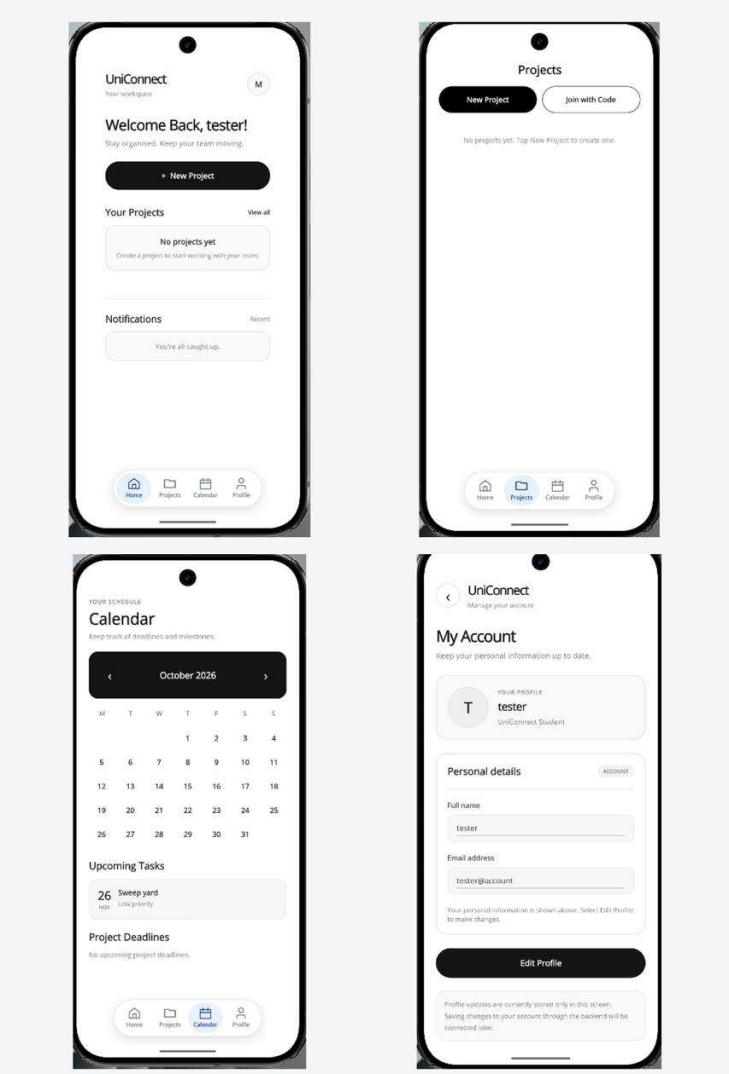
In this layer, this is the first contact, between the User and the UniConnect interface. The UniConnect app uses 1FA, which is a single factor authorisation. This involves the user setting up a password for themselves, rather than the hassle of having to answer many access questions. And because the application aims to be lightweight, we decided to stick with 1 authentication measure. After the user enters their password it is hashed using Spring Security's BCryptPasswordEncoder.  After successful account creation, the user's information is sent to our database where it is stored.  They are then led back to the home screen to log in.



## Dashboard, Projects, Calendar and My Account Screens

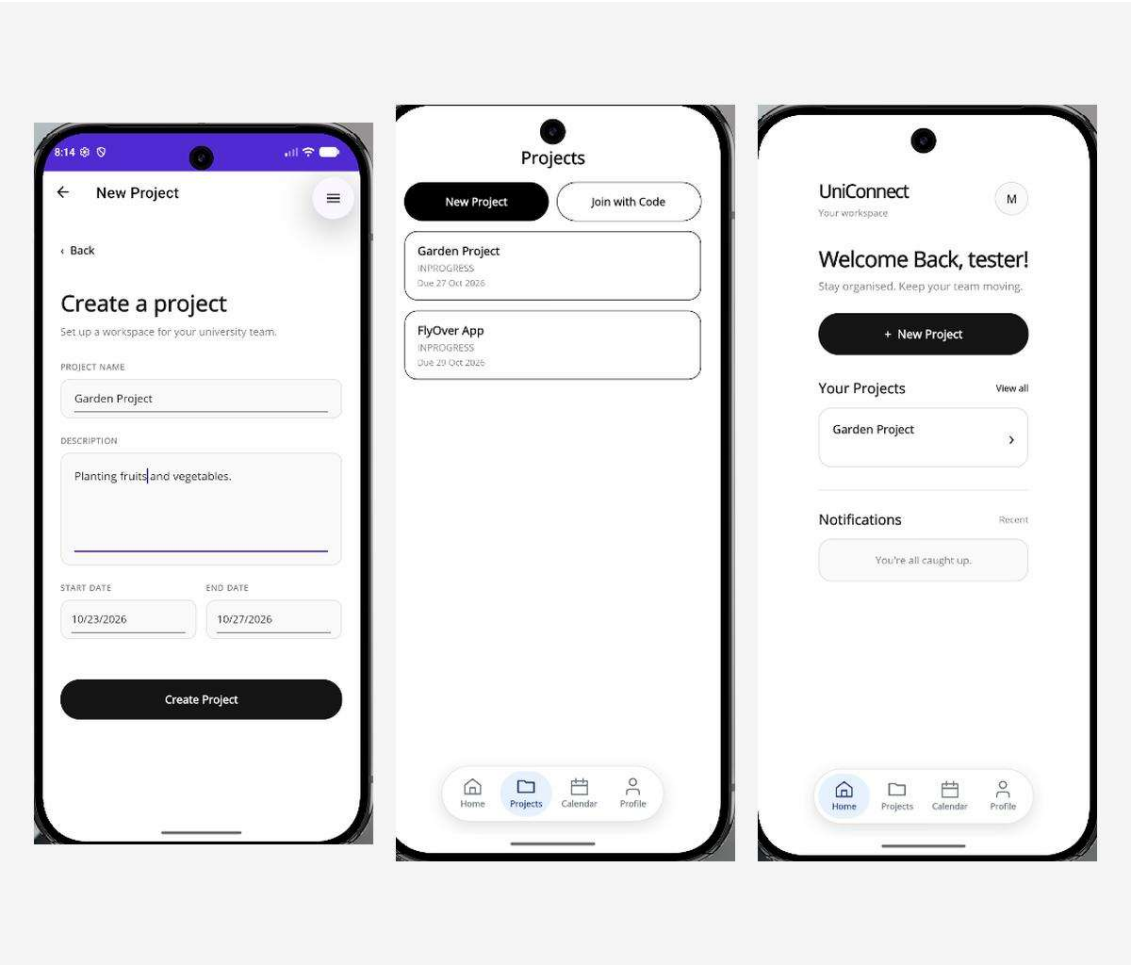
In this screen we see the dashboard, and what the dashboard does is that it provides us with a quick access to the main features of the application, while the Projects tab takes us to screen where it allows users to view and manage their projects, supporting organisation of the project and collaboration with team members. These features help users to access relevant project information and navigate through the app easily

The Calendar screen helps users to be able to keep track of important deadlines and scheduled activities, making it easier to distribute and plan their work. My Account Screen provides access to personal profile information. Together these screens will support project orgranisiation, deadline management and easy access to essential features within UniConnect.



## Project Creation

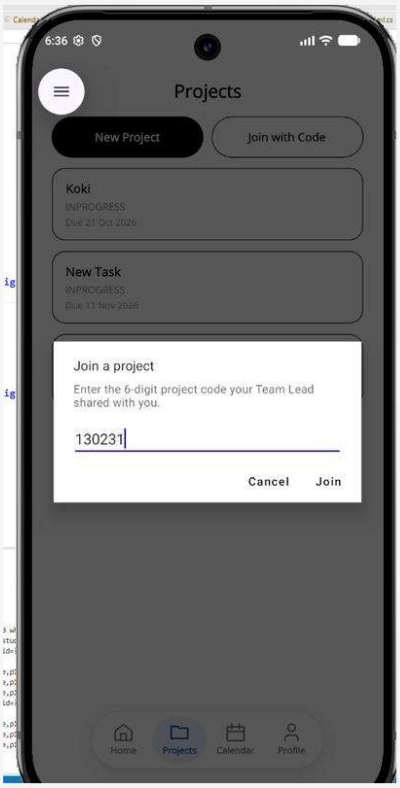
Project creation is one of the app's foundational features, in this feature, the current logged in user can create a new project, either from the button in the dashboard, or from the button in the projects page. Additionally, the current user can use a project code provided by a team leader of a group they are participating in, where they can click the "Join By Code" option and enter the code to join. All projects that the current user is involved in will appear on the projects screen.

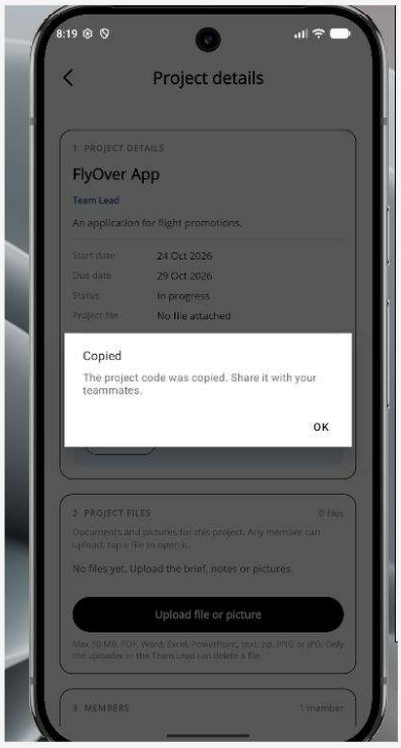


## Projects and Project Detail Screens

The screen shows an existing project and provides an option to users to create a new project or join a project that has already been created using a six-digit project code shared by the team leader.

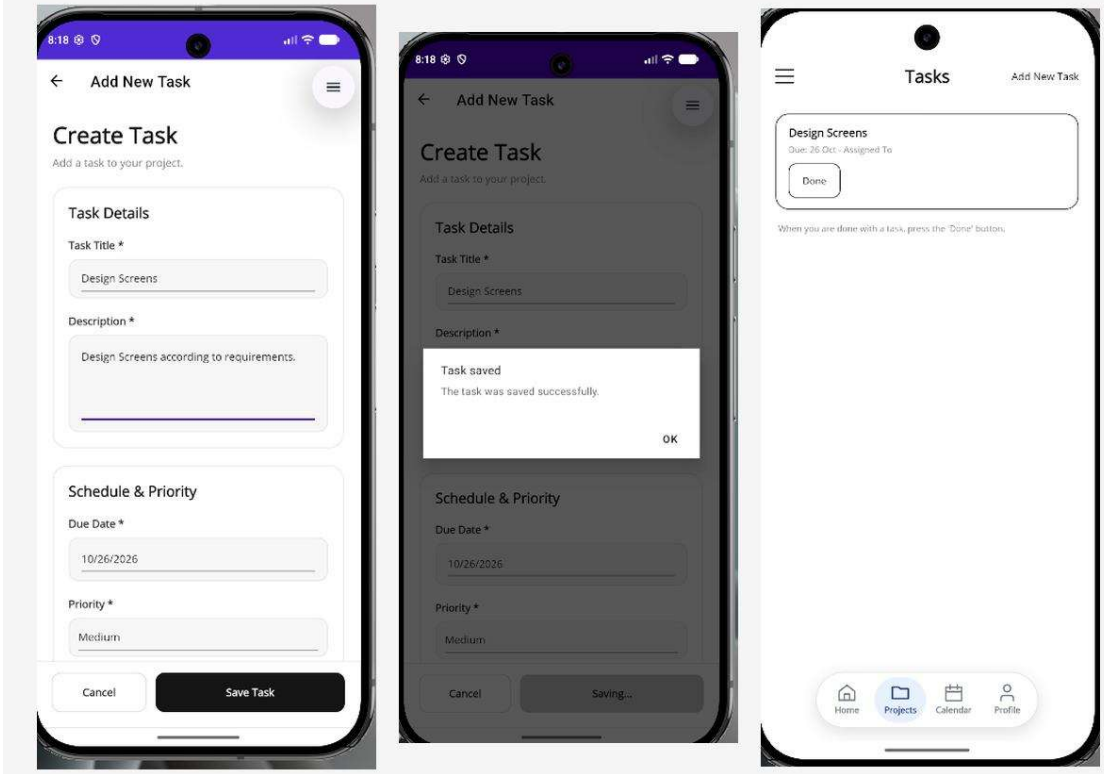
The project details screens display the name, description, team leader, start date, due date and status. It includes sections for project files and team members, while a confirmation message informs the user when the project code has been copied.



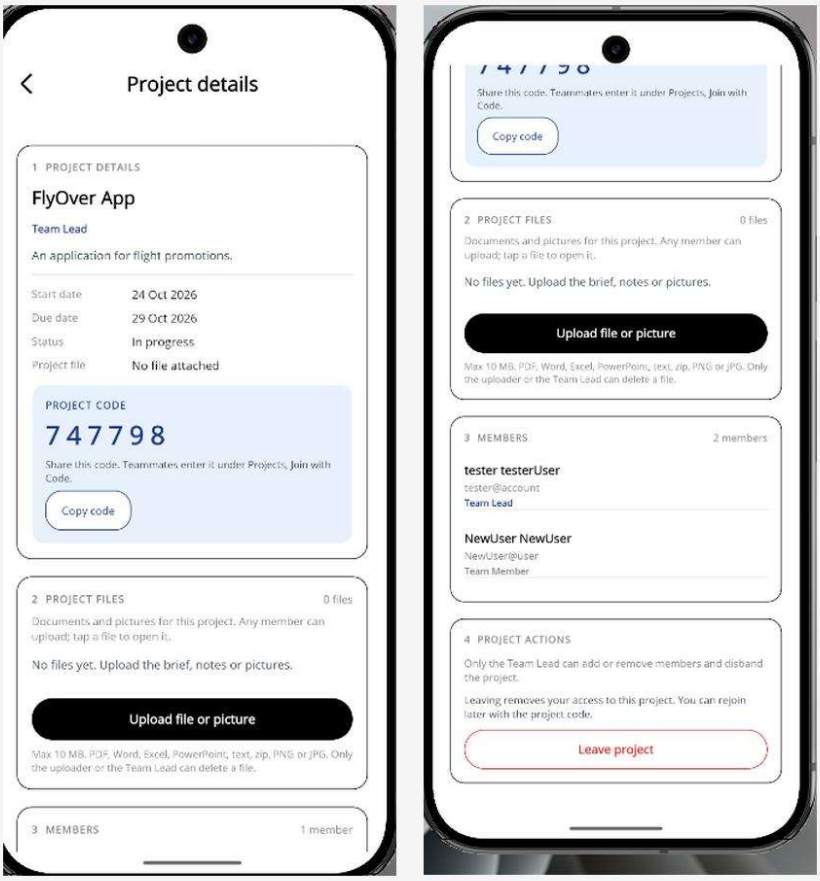
## Task Creation

Task Creation is also another foundational component of the application. Through task creation, the project can be broken down into smaller tasks and distributed. The task creation screen allows for users to give their task a title, description, a priority level, and assignation, before it is saved to the database. The tasks appear on the tasks screen which is the screen that appears first when opening a project.



## Project Details

Project details are accessed within a project. This screen provides you with a project code to use to join the project, and information on the project, mainly its members, start and end dates, and a section to upload a project file. Also, as a leader, you have the ability to disband the project. Members also have the ability to leave the project.



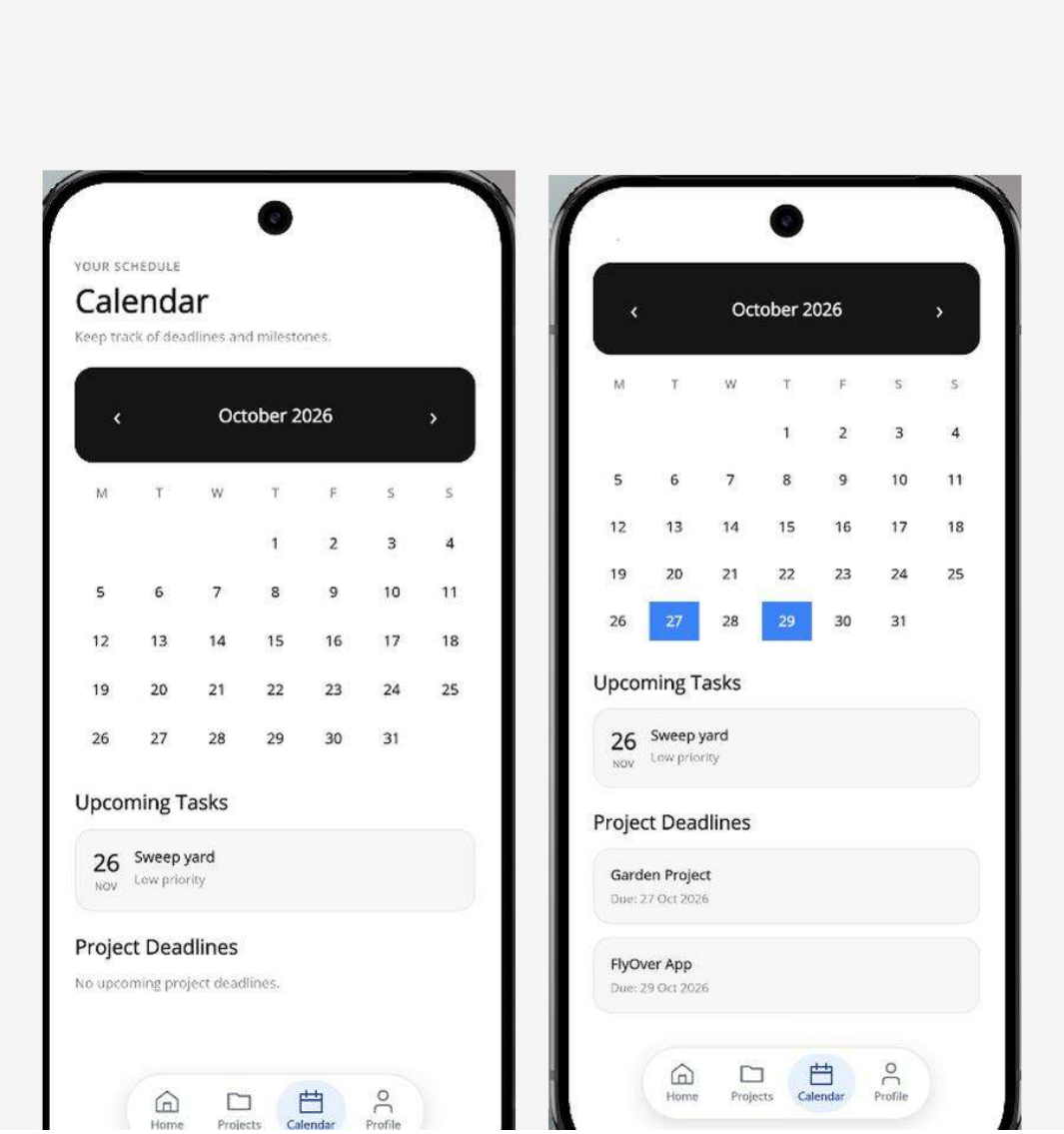
## Communication Layer

The chat is an additional feature in UniConnect. It aims to make better collaboration through communication. Each project is assigned its own dedicated chat where project members can communicate about the project and share updates. This is also a part of trying to provide an all-in-one environment for users.



## Calendar

In a way the Calendar is a big part of the communication aspect. It displays the due dates of upcoming projects and tasks, to keep the user alert.



## Sample Code of Technologies Used

In this section the snippets of the code are for the technologies used to develop the UniCnnect. These examples demonstrate how the frontend, backend and database are implemented and how they all come together to build the application.

### C# and .NET MAUI

These two are used to implement the mobile application's interface and user interaction. The snippet code that follows demonstrates how the Login screen retrives the user's credentials, calls the login function and navigates to the dashboard after it has successfully logged in

```
private async void OnLoginClicked(object sender, EventArgs e)
{
    _viewModel.Email = EmailEntry.Text;
    _viewModel.Password = PasswordEntry.Text;

    bool success = await _viewModel.LoginAsync();

    if (success)
        await Shell.Current.GoToAsync("//DashboardPage");
}
```

### Java and  Spring Boot

Spring Boot is used to implement the backend authentication endpoints for UniConnect. This specific snippet handles the user registration and login requests through the endpoints.

```
@PostMapping("/register")
    public Student registerUser (@RequestBody Student student){


        //Variables
        String name = student.getName();
        String surname = student.getSurname();
```

```
String email = student.getEmail();
String password = student.getPassword();


if (name.isEmpty()){
  throw new ResponseStatusException(HttpStatus.BAD_REQUEST,"Name field cannot be empty.Please fill it in.");
}else if (surname.isEmpty()){
  throw new ResponseStatusException(HttpStatus.BAD_REQUEST,"Surname field cannot be empty.Please fill it in.");
}else if (email.isEmpty()){
   throw new ResponseStatusException(HttpStatus.BAD_REQUEST,"Email field cannot be empty.Please fill it in.");
} else if (password.isEmpty()){
   throw new ResponseStatusException(HttpStatus.BAD_REQUEST,"Password field cannot be empty.Please fill it in.");
}else if (password.length() <= 4){
   throw new ResponseStatusException(HttpStatus.BAD_REQUEST,"Password cannot be less than 4 characters.");
}else{
   return studentService.createUser(name, surname,email, password);
}
```

MySQL Database Management

MySQL is used to store and organise UniConnect's application data. The databse supports the storing and retriving of data used by features like user accounts. XAMPP provides the local MySQL  envirnment used during dvelopment.

**Api Integration. Frontend and Backend Communication**

The ApiService class connects the UniConnect mobile apllication to the Spring Boot backend HTTP requests. It sends data to the API endpoints and recieves responses that can be used by the frontend

```
public async Task<List<TaskModel>> GetTasksAsync(long projectId)
 {
    if (projectId <= 0)
       throw new Exception("A valid project is required.");

    var response = await _http.GetAsync(
       $"/tasks/getTaskByProject/{projectId}");

    response.EnsureSuccessStatusCode();

    return await response.Content.ReadFromJsonAsync<List<TaskModel>>()
       ?? new List<TaskModel>();
 }
```

**CRUD Operations  - Task Management**

The TaskController manages task-related requests in UniConnect through REST API endpoints. It supports creating tasks, retrieving tasks by projects, updating task details and deleting tasks. These operations are handled by the TaskService class.

```
@PostMapping("/createTask/{projectId}")
   public Task createTask(@PathVariable Long projectId, @RequestBody Task task)
{


  }
```

**Data Models- Task Entity**

The Task entity represents task information in UniConnect and is mapped to a database table using Jakarta Persistence. It stores details such as the task title, description, due date, status, priority and everything associated to project ID and assigned student ID.

```
Entity

public class Task {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    @JsonProperty("taskId")
    private Long id;

    @JsonProperty("title")
    private String title;
    private String description;
    private LocalDate dueDate;
    private String status;
    private String priority;

    private Long projectId; // Which project this task belongs to


    private Long assignedTo; // the id of the student this task is assigned to

    public Task() {
    }
```

## Conclusion

Overall UniConnect is a student-centered mobile application designed to simplify project management, atsk organisation, deadline tracking and team communication. Developed using .NET MAUI, C#, Java Spring  Boot and MySQL, the project demonstrates the applicatuon of software engineering principles and structured development. Although some integrations require improvements UniConnect provides a strong foundation of the platform that helps studnets manage academic work and collaborate more efficiently.

GitHub Link: https://github.com/TSHEPANGBAM20/UniConnect.git

Demonstration Video: UniConnect Application Walkthrough

Disclaimer: To get access to the walkthrough of the major screens and features of UniConnect, please Click the link under “Demonstration Video” above, if the link doesn’t work, please refer to the link in the README.md file in the main branch of the UniConnect repository also attached above.